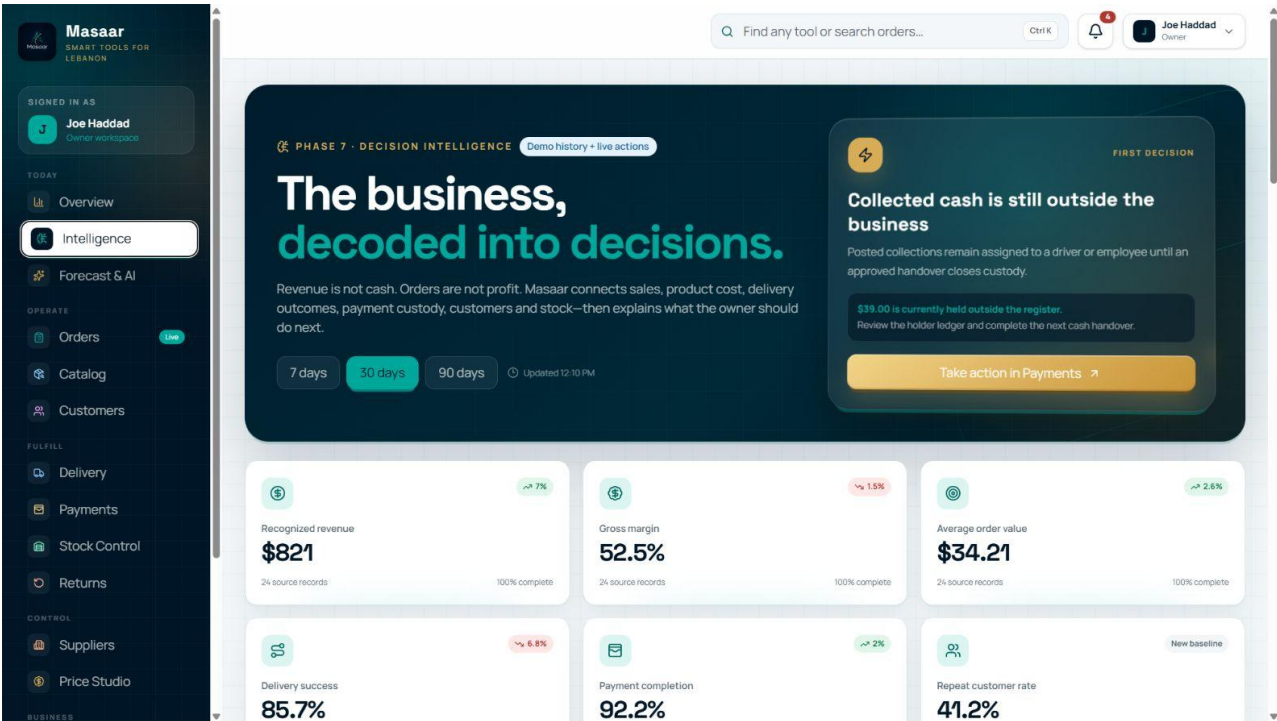


MASAAR

The operating system behind Lebanese social commerce

Orders, delivery, payments, stock and business intelligence—one accountable path after the customer decides to buy.

Competition thesis: Small businesses do not need another disconnected dashboard. They need reliable operations that produce trustworthy decisions: what is selling, what is profitable, who holds the cash, where delivery fails and what to do next.



IMPLEMENTED	VERIFIED	DEPLOYMENT TARGET
Role-based end-to-end product	55 automated tests after security hardening	AWS + Cognito + RDS SQL Server
Lebanon-first operating controls	Lint, typecheck, build and responsive review	CloudFront PWA + API Gateway + Lambda

Prepared for the Amazon Industry Program competition | Masaar Team | 31 August 2026

PROBLEM EVIDENCE

Lebanese commerce loses truth after the sale

Survey themes and field observations point to one high-value niche: the post-order operating loop.

The real operating problem

A sale may begin in Instagram or WhatsApp, but the work that follows is usually copied into chats, screenshots, spreadsheets and memory. Customer details, exact variants, addresses, delivery attempts, payments and driver-held cash diverge. Owners then cannot answer basic questions without calling several people.

Observed pressure	Business consequence	Masaar control
Fragmented social orders	Forgotten or duplicated work	Structured Quick Order + secure customer confirmation
Weak local addresses and mixed couriers	Failed attempts and disputed responsibility	Lebanese zones, landmarks, resources and immutable attempts
USD/LBP volatility and incomplete costs	Prices rise while real margin falls	FX snapshots, cost completeness and Price Studio
Cash-heavy delivery and local payments	Delivered is mistaken for paid; cash goes missing	Payment ledger, cash custody and reconciliation
Supply delays and low connectivity	Stockouts, dead stock and late updates	Reservations, supplier radar and offline-aware driver queue

Focused product boundary

Masaar does not replace Instagram, WhatsApp or a merchant’s storefront. It begins when the customer decides to order and owns the path until the parcel, payment, cash handover, stock consequence and business insight are all accountable.

CAPTURE	CONFIRM	FULLFILL	DELIVER	COLLECT	DECIDE
Product + customer	Secure address	Prepare + reserve	Assign + attempt	Pay + reconcile	Explain + act

Lebanon-specific advantage

- Original USD/LBP values, owner-approved exchange-rate snapshots and honest margin confidence.
- Cash, Whish and OMT-friendly records; driver-held money remains visible until accepted handover.
- All governorates and unlimited served areas, standardized building/landmark fields and configurable internal, freelance or company delivery resources.
- Low-connectivity mobile behavior, supplier lead-time exposure and administrative reminders without pretending to provide legal advice.

IMPLEMENTED PRODUCT

Operational truth becomes business intelligence

Every graph is downstream of a role-owned event; no decorative dashboard and no autonomous high-impact action.

What judges can operate today

Workspace	Implemented capability	Decision/value
Owner	Action Center, approvals, setup, audit and full intelligence	See the business and assign the next action.
Employee	Catalog, customers, orders, packing, payments, stock and returns	Process work without owner-level noise.
Driver	Assigned stops, call/address, amount due, attempt result and collection	One phone-first route; cash remains accountable.
Analyst	Read-only operational evidence and BI	Inspect performance without changing records.
Customer	No-account confirmation and safe tracking link	Correct the address once and follow progress securely.

Decision intelligence that changes action

Question	Masaar answer
Are sales becoming cash?	Revenue, recognized sales, collected amount, outstanding balance and cash by current holder are separate.
What truly earns?	Product/channel revenue, cost completeness and gross contribution are shown without calling incomplete data net profit.
Where is delivery leaking?	Lebanon area growth map, success rate, first-attempt rate, cost and failure-reason Pareto.
What should we stock?	Available vs reserved, velocity, lead time, restock cash and explainable recommendation.
Who will buy again?	New/repeat cohorts, segments and observed value—private reliability caution is never a blacklist.
What needs attention now?	Ranked alerts link directly to Payments, Delivery, Stock, Orders or Customers with evidence.

Governed forecast and assistant

- Forecasts are ranges with method, history window and confidence—not promises.
- Scenario planning changes assumptions only; it never silently changes real prices or inventory.
- Masaar AI answers from the authorized tenant's records, cites facts and states what it cannot know.

Critical distinction: Delivery status, payment status and physical cash custody are three different truths. This is the core accountability innovation.

ARCHITECTURAL DESIGN

Production-grade, tenant-scoped and AWS-ready

Shared contracts, explicit trust boundaries and SQL Server operational views support both application workflows and judge-visible BI.

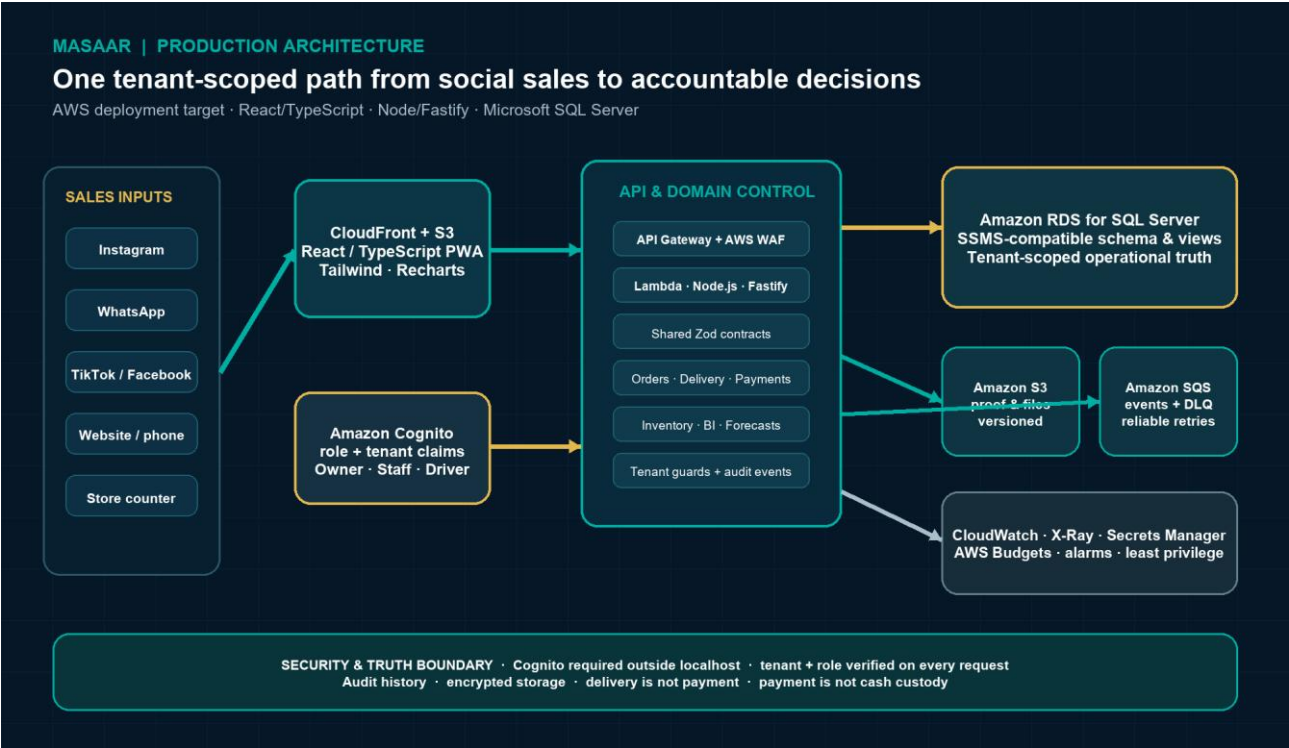


Figure 1. Masaar production architecture and security boundary.

Design choice	Why it matters
React/TypeScript + Tailwind + Recharts	One responsive PWA for owner, employee, analyst and driver; visible hierarchy without separate mobile codebases.
Node.js/Fastify + shared Zod contracts	The same schemas validate UI inputs, API payloads and tests; failures are explicit and auditable.
RDS for Microsoft SQL Server	Relational integrity, transactions and SSMS-compatible tables/views for orders, payments, stock and BI.
Cognito + tenant/role claims	Development identities are rejected in staging/production and may bind only to localhost.
CloudFront + API Gateway/WAF + Lambda	Economical managed delivery with TLS, rate limiting, tracing and a single public origin for web and API.
S3, SQS/DLQ, CloudWatch, Secrets, Budgets	Versioned evidence, durable events, observability, protected credentials and bill-shock controls.

ENGINEERING EVIDENCE

Built and verified as a system

The competition build uses production patterns while clearly separating local demo behavior from public deployment requirements.

Quality gates

Gate	Evidence
Automated contracts/API/UI	55 tests cover permissions, schema constraints, order/cash/inventory intelligence and major React workspaces.
Static quality	ESLint, TypeScript project checks and production compilation run across the monorepo.
Security hardening	Staging/production require Cognito; dev auth is loopback-only; production UI has no demo shortcuts or token fallback.
Database migration	MongoDB was removed; SQL Server schema, seeds, constraints, indexes, audit events and operator/BI views are supplied.
Responsive review	Owner, employee and driver workflows were exercised at desktop and phone dimensions.
Integration honesty	In-app workflows are live; external WhatsApp/email/courier/payment APIs remain visibly gated until approved credentials exist.

AWS release sequence

1	2	3	4	5
Provision RDS + secret	Deploy SAM stack	Publish PWA	Seed tenant + Cognito users	Smoke-test + record

Mentor feedback closed

- Hardcoded demo users remain usable only in local/test mode; configuration now fails closed outside loopback and production bundles hide role shortcuts.
- CloudFront now proxies /api/* and /health to API Gateway rather than sending application calls to S3.
- The final report now contains the requested architectural design diagram, balanced team ownership, explicit evidence and honest deployment status.

Current external gates

AWS deployment: No AWS profile, account credentials, RDS connection string, budget email or domain are present on this machine. The stack and CI workflow are ready, but a real public deployment cannot be completed safely without those account-owned values.

TEAM & HANDOFF

Five balanced ownership streams

Replace the four bracketed labels with the real member names; the contribution scope is already divided across the entire project.

Team member	One-sentence contribution
Shawn Lamouza	Led product integration, requirements, architecture decisions and competition coordination across the complete Masaar release.
[TEAM MEMBER 2]	Owned Lebanese business research, survey design/analysis, problem validation and customer/order workflow requirements.
[TEAM MEMBER 3]	Owned Figma/brand translation, React/Tailwind interface engineering, responsive behavior and role-specific usability.
[TEAM MEMBER 4]	Owned Fastify APIs, shared Zod contracts, SQL Server data integrity, AWS infrastructure, Cognito and audit/security controls.
[TEAM MEMBER 5]	Owned BI metrics/graphs, forecast logic, test validation, user documentation, demo narrative and final submission packaging.

Submission links and deliverables

Item	Submission value
Public GitHub	https://github.com/shawnlamouza/masaar repository creation/push awaiting final GitHub-side authorization
Public demo	AWS stack prepared; URL awaits account credentials, RDS and deployment.
Demo video	output/video/Masaar_2_Minute_Demo.mp4 upload link required after file creation.
Main report	output/pdf/Masaar_Final_Submission.pdf + editable Word source
Supporting package	Functional specification, implementation plan, user manual, SQL scripts, source code and clean ZIP.

Final competition statement

Why Masaar wins: Masaar is not a generic ERP and not a dashboard pasted onto unreliable data. It owns the difficult Lebanese post-order loop, protects the parcel, money and stock, and converts accountable events into practical decisions an owner can defend.

Open external items before official submission: four teammate names, GitHub repository creation/push, AWS credentials/deployment and public video upload.